Vol. 97, No. 1

NORMAL INTERPEDICULATE DISTANCES (MINIMUM AND MAXIMUM) IN CHILDREN AND ADULTS*

By VINCENT C. HINCK, M.D., WILLIAM M. CLARK, Jr., M.D., *and* CARL E. HOPKINS, Ph.D.
PORTLAND, OREGON

Downloaded from www.ajronline.org by 151.181.39.14 on 10/27/26 from IP address 151.181.39.14. Copyright ARRS. For personal use only; all rights reserved

In 1934 Elsberg and Dyke[2] published normal values for interpediculate distances as determined on roentgenograms of the adult human spine. Results were expressed as "usual upper limits" and "extreme upper limits" of normal but the method by which these limits were determined was not defined. Their use in diagnosis of intraspinal tumor was demonstrated.

In 1953 Landmesser and Heublein[6] prepared measurements of interpediculate distances using roentgenograms of selected children from the population of a home and hospital for crippled children. Data were published as ranges of normal without reference to mean values or variance.

In 1955 Simril and Thurston[9] published maximum and minimum normal interpediculate distances of infants and children. Their subjects were grouped according to "estimated physical age" by use of the Wetzel grid and their norms expressed as maximum and minimum for 3 age groups (1 day to 2 months; 2 months to 2 years; and 2 years to 12 years). Although they included minimum norms, they made no specific reference to their practical value nor did they describe the method of data analysis by which they arrived at values for maximum and minimum normal.

The following year Schwarz[7] published "extreme upper limits" of normal interpediculate distances in children and adults, adding measurements of the sacral spinal canal. To our knowledge, these have been the only attempts to establish normal ranges for interpediculate distances.

These pioneer studies have established the clinical value of the interpediculate measurements in the diagnosis of intraspinal tumors at all ages. There is need, however, for a more definitive delineation of the range of normal interpediculate distances, particularly with regard to the variability of their values in any given age group at the various spinal canal levels.

PURPOSE OF INVESTIGATION

The present investigation was undertaken to provide refined *maximum* normal interpediculate measurements for use in diagnosis of intraspinal tumor and *minimum* norms for recognition of spinal canal stenosis. Since use of interpediculate measurements in tumor diagnosis needs no endorsement, emphasis will be on diagnosis of stenosis.

METHOD

The roentgenographic files of the University of Oregon Medical School were systematically searched for anteroposterior spine roentgenograms of white patients. The roentgenograms were screened for readability and attempt was made to eliminate subjects with significant anomalies and other problems likely to influence growth and development. This selection was made to minimize bias introduced by use of a hospital population. Studies selected for measurement were classified by sex and by age to the nearest month. (Technique: target-film distance = 40 inches.) All measurements were made by one of the authors using calipers and were recorded to the nearest tenth mm. Interpediculate distance (hereinafter IPD) was considered the shortest measurement between the medial surfaces of the pedicles of a given vertebra.

IPD measurements were made on 474 roentgenograms, including 353 of children

* From the Departments of Radiology and Pediatrics of the University of Oregon Medical School, Portland; and the School of Public Health, University of California, Los Angeles, California.
Supported by the Easter Seal Research Foundation of the National Society for Crippled Children and Adults.

TABLE IA

MEAN INTERPEDICULATE DISTANCE (IPD) OF EACH VERTEBRA BY AGE

Male and Female Combined (mm.)

Age Group (yr.)

	3, 4, 5	6, 7, 8	9, 10	11, 12	13, 14	15, 16	17, 18	Adult
Age (yr.)	4.1	7.1	9.4	11.6	13.6	15.6	17.5	>18
Vertebra								
C3	23.9	26.0	26.4	26.2	27.2	27.1	27.7	28.0
4	24.9	26.8	26.9	27.0	28.3	28.2	28.7	28.8
5	25.3	27.2	27.1	27.4	28.6	28.6	29.1	29.4
6	25.3	27.6	27.0	27.3	28.5	28.2	29.1	29.3
7	24.4	26.4	26.2	25.9	27.2	26.9	27.6	28.0
T1	21.4	22.4	23.4	23.5	23.4	23.5	23.1	24.0
2	18.2	18.9	20.0	19.8	20.1	19.9	19.8	20.5
3	16.9	17.6	18.4	18.1	18.7	18.1	18.4	18.8
4	16.0	17.0	17.6	17.2	18.1	17.3	17.8	18.1
5	15.8	16.8	16.9	16.8	17.7	17.1	17.5	17.6
6	15.8	16.7	16.5	16.8	17.8	16.7	17.3	17.3
7	16.0	16.9	16.5	16.8	18.2	16.9	17.5	17.4
8	16.3	17.6	16.7	17.1	18.4	17.3	18.0	17.7
9	16.4	17.8	17.0	17.4	18.7	17.7	18.3	18.2
10	16.4	18.1	17.2	17.7	18.8	17.8	18.8	18.7
11	17.6	19.2	18.5	18.9	20.3	19.5	20.0	20.2
12	19.9	21.2	21.2	21.4	23.1	22.3	22.8	23.2
L1	20.5	21.8	23.0	23.0	23.7	23.9	24.5	25.0
2	20.4	21.9	23.2	23.1	23.4	23.9	24.4	25.5
3	20.8	22.4	23.8	23.6	24.0	24.6	24.8	26.0
4	21.7	23.3	24.8	24.9	25.7	25.6	26.1	26.9
5	24.5	26.5	28.3	28.7	29.1	29.1	29.6	29.7

(under the age of 19 years) and 121 of adults. The first 2 cervical vertebrae were excluded as unmeasurable and sacral measurements were omitted for want of an adequate sample. Since the roentgenograms had been taken for purposes other than this study, it was not possible in all cases to visualize the entire spine for measurement. Hence, the sample size varies from vertebra to vertebra. In general, the lumbar segment is most heavily represented and the cervical the least.

MEASUREMENT ERROR

Eighty-nine duplicate measurements were made, independently and at different times, in 9 different cases. The differences between duplicates ranged from zero to an extreme of 3.6 mm. The median absolute error was 0.26 mm. and, because of a few extremes, the mean difference was 0.40 mm. Of the 89 duplicates measured, 10 per cent had zero error, 50 per cent had less than 0.26 mm., 75 per cent had less than 0.50 mm. and 90 per cent had less than 1.0 mm. error.

RESULTS

IPD mean values and standard deviations were computed for each vertebral level separately by sex and for broad age groupings (3, 4, 5; 6, 7, 8; 9, 10; 11, 12; 13, 14; 15, 16; 17, 18 years and adult). Age group mean values are given disregarding sex in Table 1A, for males only in Table 1B and for females only in Table 1C. Standard deviations and sample sizes for each age group are given in Table 11, male and fe-

Downloaded from www.ajronline.org by 151.181.39.14 on 10/27/26 from IP address 151.181.39.14. Copyright ARRS. For personal use only; all rights reserved

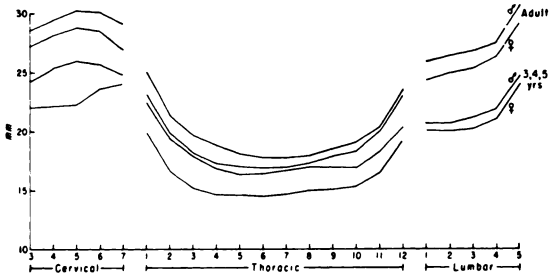


FIG. 1. Mean interpediculate distance (IPD), each vertebra, age 3, 4, 5 years and adult, by sex.

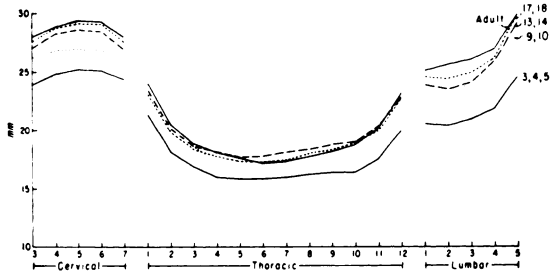


FIG. 2. Mean interpediculate distance (IPD), each vertebra, 5 age groups, sexes combined.

male combined, corresponding to the mean values in Table 1A.

The relative importance of age and sex in determining mean IPD can be seen in Table 1, or more easily in Figure 1, in which IPD mean values are plotted for each sex and for the two age extremes, 3, 4, 5 years and adult. It is evident that average male measurements are consistently larger than female by about 1 mm. The age difference is more striking, the growth being about 3 to 5 mm. from age 3, 4, 5 to adulthood. Some irregularities in the age-sex curves, *e.g.*, cervical for females age 3, 4, 5, are due

TABLE 1B

MEAN INTERPEDICULATE DISTANCE (IPD) OF EACH VERTEBRA BY AGE

Male (mm.)
Age Group (yr.)

	3, 4, 5	6, 7, 8	9, 10	11, 12	13, 14	15, 16	17, 18	Adult
Age (yr.)	4.0	7.0	9.4	11.6	13.5	15.6	17.4	>18
Vertebra								
C3	24.3	26.3	26.9	26.0	26.9	27.9	28.6	28.6
4	25.5	27.4	27.3	27.8	28.0	28.8	29.5	29.5
5	26.1	27.8	27.5	27.0	28.4	29.3	29.9	30.3
6	25.8	28.2	27.3	27.2	28.4	29.2	30.1	30.2
7	24.8	27.1	26.2	26.0	27.3	28.0	28.8	29.3
T1	22.5	22.7	23.8	23.8	24.2	25.2	24.5	25.1
2	19.4	19.7	20.4	20.6	20.6	21.7	21.2	21.4
3	17.9	18.1	19.0	18.8	19.1	19.2	19.4	19.6
4	16.9	17.6	18.2	17.6	18.3	18.0	18.7	18.8
5	16.4	17.3	17.4	17.4	18.3	17.7	18.4	18.2
6	16.5	17.3	18.8	17.4	18.3	17.3	18.2	17.8
7	16.7	17.5	16.5	17.4	18.5	17.6	17.9	17.8
8	17.0	18.1	16.7	17.6	18.9	18.3	18.4	18.0
9	17.0	18.5	17.0	17.8	19.3	18.6	18.7	18.6
10	16.9	18.8	17.3	18.1	19.5	18.1	18.9	19.1
11	18.2	20.1	18.8	19.2	20.9	19.8	20.2	20.4
12	20.4	22.5	21.3	21.8	23.5	23.2	23.4	23.5
L1	20.7	22.5	23.3	23.9	23.8	24.5	25.1	25.9
2	20.7	22.4	23.5	24.2	23.3	24.6	24.8	26.5
3	21.2	23.0	24.1	24.6	23.6	25.1	25.2	26.8
4	21.9	23.6	24.8	23.6	24.7	26.0	26.6	27.6
5	24.7	26.9	28.4	28.9	28.0	30.1	29.9	30.7

Downloaded from www.ajronline.org by 151.181.39.14 on 10/27/26 from IP address 151.181.39.14. Copyright ARRS. For personal use only; all rights reserved

TABLE IC

MEAN INTERPEDICULATE DISTANCE (IPD) OF EACH VERTEBRA BY AGE

Female (mm.)
Age Group (yr.)

	3, 4, 5	6, 7, 8	9, 10	11, 12	13, 14	15, 16	17, 18	Adult
Age (yr.)	4.2	7.2	9.6	11.6	13.7	15.5	17.6	>18
Vertebra								
C_3	22.1	25.5	24.8	26.6	27.5	26.6	26.6	27.4
4	22.2	26.0	25.6	27.4	28.4	27.7	27.5	28.2
5	22.3	26.4	25.9	28.0	28.7	28.1	27.9	28.7
6	23.6	26.7	26.0	27.3	28.6	27.5	28.0	28.6
7	24.0	25.5	26.3	25.8	27.1	26.1	26.1	27.1
T1	19.8	22.1	23.1	23.0	22.6	22.3	22.4	23.1
2	16.6	18.2	19.7	18.8	19.6	18.9	19.1	19.8
3	15.2	17.0	17.9	17.5	18.3	17.6	17.9	18.2
4	14.6	16.3	17.1	16.6	17.8	16.9	17.4	17.4
5	14.6	16.3	16.6	16.0	17.2	16.8	17.1	17.1
6	14.5	16.1	16.3	15.9	17.4	16.5	16.9	16.9
7	14.7	16.3	16.5	16.1	17.8	16.6	17.4	17.0
8	15.0	16.9	16.6	16.4	18.0	16.8	17.8	17.4
9	15.1	17.1	16.9	16.8	18.1	17.3	18.2	17.9
10	15.4	17.3	17.1	17.1	18.2	17.7	18.7	18.4
11	16.5	18.0	18.2	18.4	19.8	19.2	19.9	20.0
12	19.1	20.2	21.1	20.9	22.8	21.7	22.5	22.9
L1	20.1	21.0	22.5	22.2	23.7	23.6	24.1	24.3
2	20.0	21.1	22.6	22.3	23.6	23.6	24.2	24.9
3	20.2	21.7	23.2	22.8	24.6	24.4	24.5	25.4
4	21.1	23.0	24.7	24.2	26.9	25.4	25.8	26.4
5	23.9	26.1	28.2	28.5	30.4	28.6	29.5	29.0

to different sample sizes obtained in this refined breakdown. Since the sex effect is an order of magnitude less than that of age or clinical significance, it was decided to present the "normal" tolerance ranges irrespective of sex.

The effect of age, with sexes combined, can be seen in Figure 2. The mean values for ages 3, 4, 5; 9, 10; 13, 14; 17, 18 years and adult are plotted. There is a steady progression of mean IPD of all levels with age. In the cervical segment there is a substantial increase from age 3, 4, 5 to age 9, 10 (about 2 mm.) and another increase (1.5 mm.) to age 13, 14; thereafter, there is slight increase. In the thoracic segment the main growth again takes place in the early years from 3, 4, 5 to 9, 10 and this early growth is still more marked in the lumbar segment, the average increase from the 3, 4, 5 to 9, 10 age group being nearly 3 mm.

The interperson variability, represented by the standard deviation (Table 11), is reasonably constant for age and sex but varies considerably by vertebra. The cervical vertebrae are about equally variable, having standard deviations of 1.4 to 2.8 mm. There appears to be a slight increase in variability in the upper ages, attributable perhaps, in part, to the increased difficulty of visualization, as well as to the greater differences in general somatic size. The age group 11, 12 years was consistently more variable than any other age. This increased variability comes mainly from males who have standard deviations of 3 mm. and more in this age group. Since this

Downloaded from www.ajronline.org by 151.181.39.14 on 10/27/26 from IP address 151.181.39.14. Copyright ARRS. For personal use only; all rights reserved

Downloaded from www.ajronline.org by 151.181.39.14 on 10/27/26 from IP address 151.181.39.14. Copyright ARRS. For personal use only; all rights reserved

is the period of onset of the skeletal growth spurt for some boys, it is assumed that the increased variability is reflecting individual differences in the timing of this growth.

Interperson variability in the thoracic segment is least in the vertebrae with smaller IPD measurements (3–10) and uniform across the ages, with the exception of a slight increase in variability in the 13, 14 age group, related most probably to individual differences in growth at this period. The vertebrae with larger measurements (1, 2, 11, 12) are slightly more variable.

In the lumbar segment, there is a slight trend of increasing variability with age, but the most striking feature is a uniformly greater variability in L4 and 5 than in 1, 2 and 3. This is due to the fact that although the IPD's of L4 and 5 are generally larger than those of L1, 2, and 3, in a few instances they are actually smaller.

Norms for use in clinical appraisal have been constructed by computing, for each vertebra and each age group, 90 per cent tolerance ranges based on the combined male and female samples (Table III, Fig. 3, *A–H*). These tolerance ranges are the low and high limits within which the central 90 per cent of "normals" may be expected to fall:[1] Only 5 per cent of "normals" should be below the low limit and only 5 per cent above the high limit. The width of these ranges varies somewhat from age group to age group and from vertebra to vertebra because of 2 relevant factors: the

TABLE II

STANDARD DEVIATION OF INTERPEDICULATE DISTANCE (IPD) OF EACH VERTEBRA BY AGE

Male and Female Combined (mm.)

Age Group (yr.)

	3, 4, 5	6, 7, 8	9, 10	11, 12	13, 14	15, 16	17, 18	Adult
	Standard Deviation in mm. (Sample Size, N)							
Age (yr.)	.09 (35)	.08 (52)	.05 (30)	.05 (46)	.05 (59)	.05 (63)	.05 (68)	— (121)
Vertebra								
C3	1.7 (8)	1.5 (18)	1.7 (8)	2.6 (18)	1.2 (22)	1.8 (17)	1.9 (21)	1.5 (36)
4	1.7 (8)	1.5 (18)	1.3 (8)	2.5 (18)	1.5 (23)	1.6 (18)	2.0 (22)	1.5 (43)
5	1.7 (8)	1.7 (18)	1.6 (8)	2.6 (19)	1.7 (23)	1.6 (18)	2.1 (22)	1.7 (44)
6	1.7 (9)	1.7 (18)	1.7 (8)	2.7 (20)	2.1 (23)	1.8 (20)	2.2 (24)	1.8 (50)
7	1.4 (9)	2.0 (19)	2.1 (8)	2.8 (21)	1.8 (25)	2.4 (21)	2.1 (24)	2.0 (60)
T1	2.5 (12)	1.9 (14)	1.4 (10)	2.2 (22)	1.5 (17)	2.3 (19)	1.6 (27)	1.8 (53)
2	2.2 (12)	1.6 (14)	1.2 (10)	2.0 (23)	1.8 (17)	2.4 (21)	1.4 (27)	1.7 (53)
3	1.6 (12)	1.5 (14)	0.9 (9)	1.6 (23)	1.6 (17)	1.5 (20)	1.3 (27)	1.5 (50)
4	1.5 (12)	1.4 (14)	1.0 (9)	1.7 (23)	1.8 (15)	1.3 (19)	1.3 (26)	1.6 (43)
5	1.4 (13)	1.2 (14)	0.8 (9)	1.6 (22)	1.8 (15)	1.4 (19)	1.3 (25)	1.7 (42)
6	1.6 (13)	1.2 (13)	1.0 (9)	1.5 (22)	1.8 (15)	1.3 (19)	1.6 (25)	1.5 (42)
7	1.6 (12)	1.2 (13)	1.3 (9)	1.4 (22)	1.9 (15)	1.6 (19)	1.5 (26)	1.5 (42)
8	1.6 (13)	1.4 (14)	1.0 (9)	1.5 (22)	1.9 (15)	1.6 (19)	1.5 (25)	1.5 (41)
9	1.5 (13)	1.3 (15)	1.3 (10)	1.6 (23)	1.8 (15)	1.8 (19)	1.6 (26)	1.5 (43)
10	1.3 (14)	1.4 (17)	1.6 (10)	1.7 (23)	2.0 (17)	1.6 (20)	1.5 (26)	1.6 (44)
11	1.5 (15)	1.9 (17)	1.8 (10)	1.8 (23)	2.3 (18)	1.7 (24)	1.7 (30)	1.8 (53)
12	1.5 (17)	2.1 (17)	1.6 (11)	1.8 (24)	2.3 (20)	1.7 (28)	1.7 (31)	2.0 (59)
L1	1.6 (26)	1.9 (33)	1.8 (20)	1.7 (26)	1.6 (35)	2.0 (40)	2.4 (37)	2.2 (59)
2	1.5 (26)	2.0 (32)	1.6 (20)	1.8 (25)	1.6 (35)	1.8 (40)	2.2 (36)	2.3 (57)
3	1.6 (26)	2.1 (33)	1.7 (20)	1.9 (25)	2.0 (35)	1.9 (40)	2.3 (36)	2.7 (53)
4	1.7 (26)	2.1 (33)	2.3 (20)	3.1 (24)	3.3 (34)	2.4 (39)	2.9 (36)	3.0 (52)
5	1.9 (26)	3.1 (33)	2.7 (20)	3.1 (23)	3.6 (34)	2.9 (38)	3.6 (35)	3.7 (50)

TABLE III

TOLERANCE RANGE (90 PER CENT) OF INTERPEDICULATE DISTANCE (IPD) OF EACH VERTEBRA BY AGE

Male and Female Combined (mm.)

Age Group (yr.)

	3, 4, 5	6, 7, 8	9, 10	11, 12	13, 14	15, 16	17, 18	Adult
Vertebra								
C3	18–29	22–30	21–32	20–32	24–31	23–31	23–32	25–31
4	19–30	23–31	21–32	21–33	25–32	24–32	24–33	26–32
5	20–31	23–31	22–32	21–33	25–32	25–32	25–34	26–33
6	20–31	24–31	22–32	21–33	25–32	24–33	25–34	26–33
7	19–30	23–31	21–32	20–32	24–31	21–32	23–32	24–32
T1	17–26	19–26	20–27	20–27	19–28	18–29	20–26	20–28
2	14–22	15–22	17–24	16–24	16–24	14–25	17–23	17–24
3	13–21	14–21	15–21	14–22	15–23	15–22	15–21	16–22
4	12–20	14–21	15–21	14–21	14–22	14–20	15–21	15–21
5	12–20	13–20	14–20	13–21	14–22	14–21	15–21	14–21
6	12–20	13–20	14–20	13–20	14–22	13–20	14–20	14–20
7	12–20	13–21	14–20	13–20	14–22	13–21	15–21	14–20
8	12–21	14–21	14–20	13–21	14–23	14–21	15–21	15–21
9	12–21	14–21	13–21	14–21	15–23	14–22	15–21	15–21
10	12–21	15–22	13–21	14–21	15–23	14–22	16–22	16–22
11	13–22	16–23	14–23	15–22	16–25	16–23	17–23	17–24
12	16–24	18–25	17–25	18–25	19–27	18–26	20–26	19–27
L1	17–24	17–27	19–28	19–27	20–27	20–28	20–29	21–29
2	17–24	17–27	19–28	19–27	20–27	20–28	20–29	21–30
3	17–24	17–27	19–28	20–27	21–28	21–29	20–29	21–31
4	18–25	18–28	20–29	20–28	19–33	21–30	19–33	21–33
5	21–28	22–32	24–33	24–34	22–36	23–35	23–37	23–37

intrinsic interperson variability (estimated by the standard deviation of the observed sample) and the sample size on which our estimates of mean and variability are based. For example, the limits for L4 and L5 at age 13, 14 are quite far apart because the variation in our sample was large and, in addition, our sample sizes were smaller.

Averages of sample sizes and standard deviations of groups of vertebrae that were similar were used in computing tolerance limits by adding to and subtracting from the mean 2.8, 2.5, 2.3 and 2.1 standard deviations for estimates based on sample sizes of 10, 15, 20 and 30, respectively, following the probability tables given by Dixon and Massey.[1]

The tolerance ranges are recommended for clinical use as an index of suspicion of abnormality. Any carefully measured individual IPD falling outside the given limits should be viewed with suspicion of pathology or anomaly, though these limits are surely not stringent enough to detect all anomalies. Similarly, failure to exceed the limits cannot guarantee normalcy.

All normal spines tend to exhibit a similar profile of the 22 vertebral IPD's, individual differences being more in the overall size level than in the sizes of single vertebrae. Thus, a case in which most of the vertebrae have IPD's close to the mean value but a single vertebra or adjacent pair of vertebrae is distinctly out of line should be viewed with suspicion even though none of the individual values exceeds the tolerance limits. This type of abnormality may be suspected by comparing the *shape* of the patient's IPD curve with the normal.

Downloaded from www.ajronline.org by 151.181.39.14 on 10/27/26 from IP address 151.181.39.14. Copyright ARRS. For personal use only; all rights reserved

Downloaded from www.ajronline.org by 151.181.39.14 on 10/27/26 from IP address 151.181.39.14. Copyright ARRS. For personal use only; all rights reserved

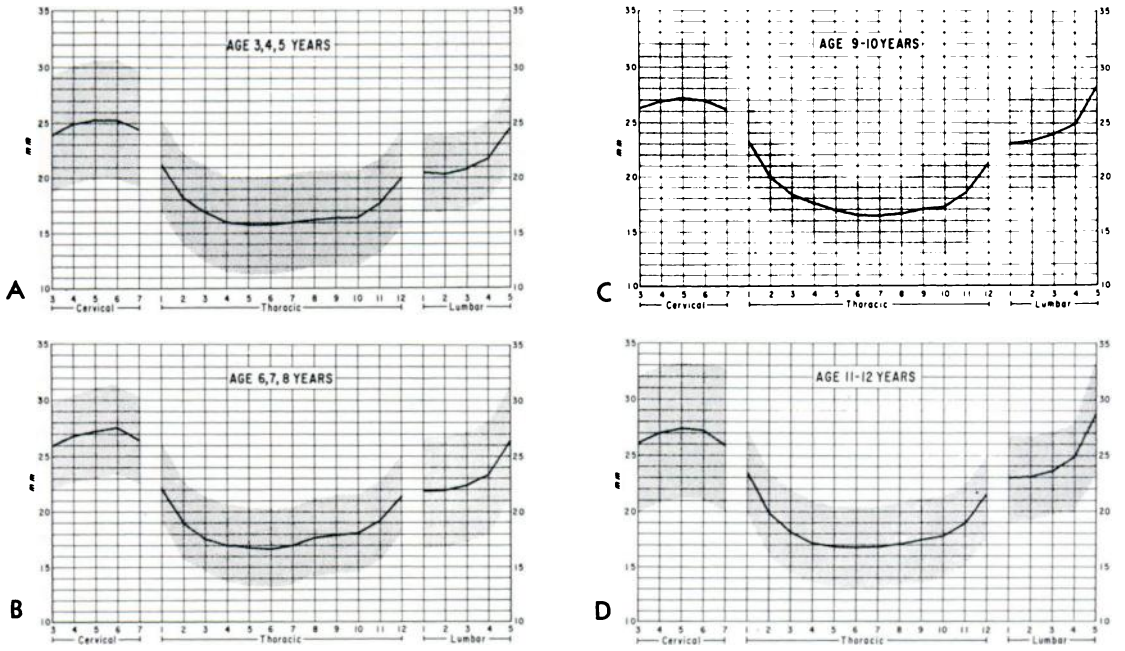


Fig. 3. Tolerance range (90 per cent) of interpediculate distance (IPD), each vertebra, male and female combined. (*A*) Age 3, 4 and 5 years. (*B*) Age 6, 7 and 8 years. (*C*) Age 9 and 10 years. (*D*) Age 11 and 12 years.

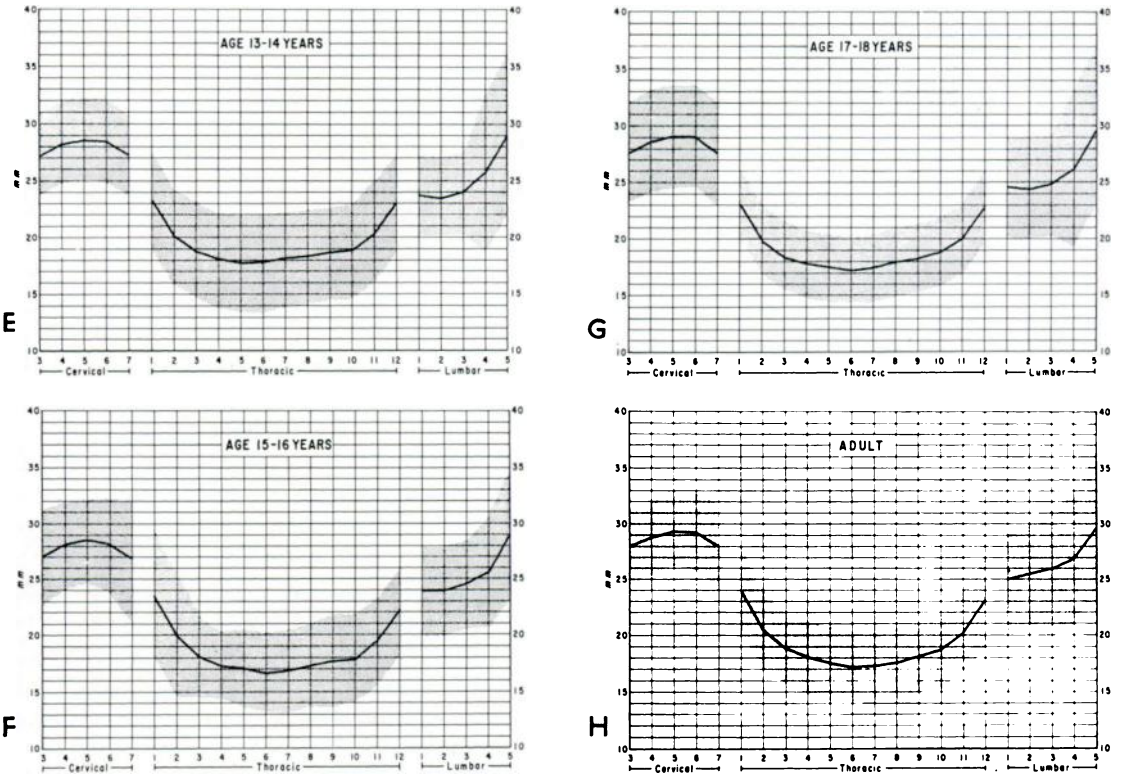


Fig. 3. Tolerance range (90 per cent) of interpediculate distance (IPD), each vertebra, male and female combined. (*E*) Age 13 and 14 years. (*F*) Age 15 and 16 years. (*G*) Age 17 and 18 years. (*H*) Adult.

Downloaded from www.ajronline.org by 151.181.39.14 on 10/27/26 from IP address 151.181.39.14. Copyright ARRS. For personal use only; all rights reserved

TABLE IV

ANTEROPOSTERIOR SPINE IPD—POSTEROANTERIOR CHEST IPD

Mean Difference at Each Vertebral Level (mm.)

Vertebra	N	Difference
T1	8	1.46
2	8	1.06
3	8	0.79
4	9	0.58
5	10	0.38
6	9	0.08
7	8	0.31
8	7	0.64
9	5	0.22
10	8	0.20
11	6	0.58
12	3	0.43

IPD: ANTEROPOSTERIOR SPINE VERSUS POSTEROANTERIOR CHEST ROENTGENOGRAMS

In clinical practice, the question occasionally arises whether IPD measurements from a posteroanterior chest roentgenogram can be compared with normal standards derived from anteroposterior spine roentgenograms. Consensus seems to have been that the 2 cannot be used interchangeably because the IPD from the chest film would be considerably magnified. To the best of our knowledge, however, this is an opinion without documentation.

Accordingly, we undertook a comparison of the 2 measurements by obtaining both posteroanterior chest and anteroposterior thoracic spine roentgenograms (target-film distances 72 inches and 40 inches, respectively) of each of 10 randomly selected normal patients between 10 and 15 years of age. IPD measurements were made on both roentgenograms in the usual manner. Due to obscurity of some vertebrae on several chest roentgenograms, there were 89 comparable measurements instead of 120. To our surprise IPD measurements from the spine roentgenograms were *larger* than those from the chest roentgenograms. (Silverman[8] independently has made this observation.) On the average the difference was about 0.6 mm.

The data were further analyzed to determine whether there was local variation in this difference. The mean difference between the 2 measurements was determined for each vertebra. The results are shown in Table IV and illustrated in Figure 4. It will be noted that there *are* local variations of the mean difference; the difference is more striking when values for T1–3 (mean difference of 1.10 mm.) are compared with those of T4–12 (mean difference of 0.38 mm.).

It may be that variations of IPD difference at these 2 levels are due to differences of relationship between spine and cassette in the 2 projections. When the individual is positioned for an anteroposterior recumbent thoracic spine roentgenogram, the vertebral column tends to straighten but still retains some of its kyphotic curve. T1

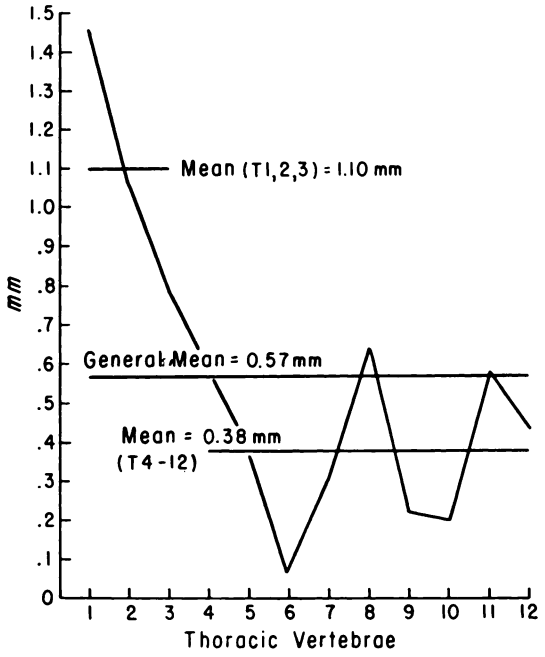


FIG. 4. Mean difference between thoracic IPD measurements from anteroposterior thoracic spine roentgenograms and those from posteroanterior chest roentgenograms; recorded for individual vertebrae, for subdivisions of the thoracic spine, and for the thoracic spine generally.

Downloaded from www.ajronline.org by 151.181.39.14 on 10/27/26 from IP address 151.181.39.14. Copyright ARRS. For personal use only; all rights reserved

is farther from the film than T6 and its IPD is more magnified than that of T6. Conversely, when positioned for an erect chest roentgenogram, the shoulders are hunched forward toward the cassette and T1 is closest to the cassette; T6 is farthest away.

Since anteroposterior measurements are generally greater than posteroanterior measurements (*i.e.*, more magnified) and T1 is the most magnified vertebra in the anteroposterior roentgenogram and the least magnified in the posteroanterior roentgenogram, it follows that the anteroposterior-posteroanterior difference should be larger for T1 than for the other thoracic vertebrae. Conversely, the anteroposterior-posteroanterior difference for T6 should be the smallest. Vertebrae between T1 and T6 should show a gradient of diminishing difference. Below T6, where the spine curve reverses and enters the "small of the back," the trend can be expected mildly to reverse. The curve of mean differences (anteroposterior spine–posteroanterior chest, Fig. 4) does indeed reflect these expected variations.

Though chest roentgenograms are *not* recommended for IPD determinations, it would appear from the foregoing analysis that they can be used if necessary with moderate accuracy, particularly for IPD's below T3.

DISCUSSION

Stenosis of the spinal canal due to decreased IPD is, to the best of our knowledge, virtually unexplored. Stenosis due to decreased sagittal diameter has been reported in the cervical spine[5] as well as the lumbar spine.[3,4,10] In either instance the anomaly may be clinically significant.

We undertook the determination of normal minimum IPD standards as a preliminary to clinical investigation of transverse spinal canal stenosis, its frequency and its significance. Our clinical experience is still limited but we can already state that (a) there is such an entity as developmental stenosis of the spinal canal with decreased IPD, (b) it may, but need not be, associated with decreased sagittal diameter, and (c) it may be symptomatic. By way of illustration the following cases are presented.

ILLUSTRATIVE CASES

CASE 1. R.C. was a 60 year old male. Eleven months before admission, he first noted abdominal swelling which persisted until the time of admission. Six months before admission, he developed severe lancinating pains in the right upper quadrant and similar pains shooting into both legs. The pains lasted several seconds and then disappeared completely. They occurred as often as 10 times a day, in any position, day or night. He was studied at another hospital where he was thought to have a "right flank hernia" and was found to have diabetes mellitus that was easily controlled by appropriate diet. The symptoms continued. Three months before admission, unsteadiness of gait and weakness of legs developed, causing difficulty in climbing stairs. He also complained of mild paresthesias of the soles of the feet.

On examination, weak and flaccid abdominal musculature was found. This accounted for the apparent abdominal enlargement. Reflexes in the upper extremities were normal. Deep tendon reflexes were absent at both ankles and the right knee; and reduced at the left knee. Babinski responses were flexor. Cremasteric reflex was absent on the right. Abdominal reflexes were absent. Romberg's sign was negative. Position and vibratory sensations were moderately decreased in the lower extremities. There was patchy and variable decrease in temperature, light touch and superficial pain sensations in the lower extremities. Examination of the spinal fluid was normal except for a protein of 84 mg. per cent.

Roentgen Findings. Thoracic spine roentgenograms showed mild degenerative changes in the middle and lower thoracic spine and a small posterior osteophyte at T7–8. Although there were no available standards for minimal IPD's, it was believed that there was, at least, a relative stenosis of the mid-thoracic spinal canal. Due to minimal rotation of the lateral roentgenogram, the sagittal diameter could not be viewed. Morphology of vertebrae was otherwise unremarkable. The IPD measurements have subsequently been compared with our normal standards and are shown in Figure 5 *B*

Downloaded from www.ajronline.org by 151.181.39.14 on 10/27/26 from IP address 151.181.39.14. Copyright ARRS. For personal use only; all rights reserved

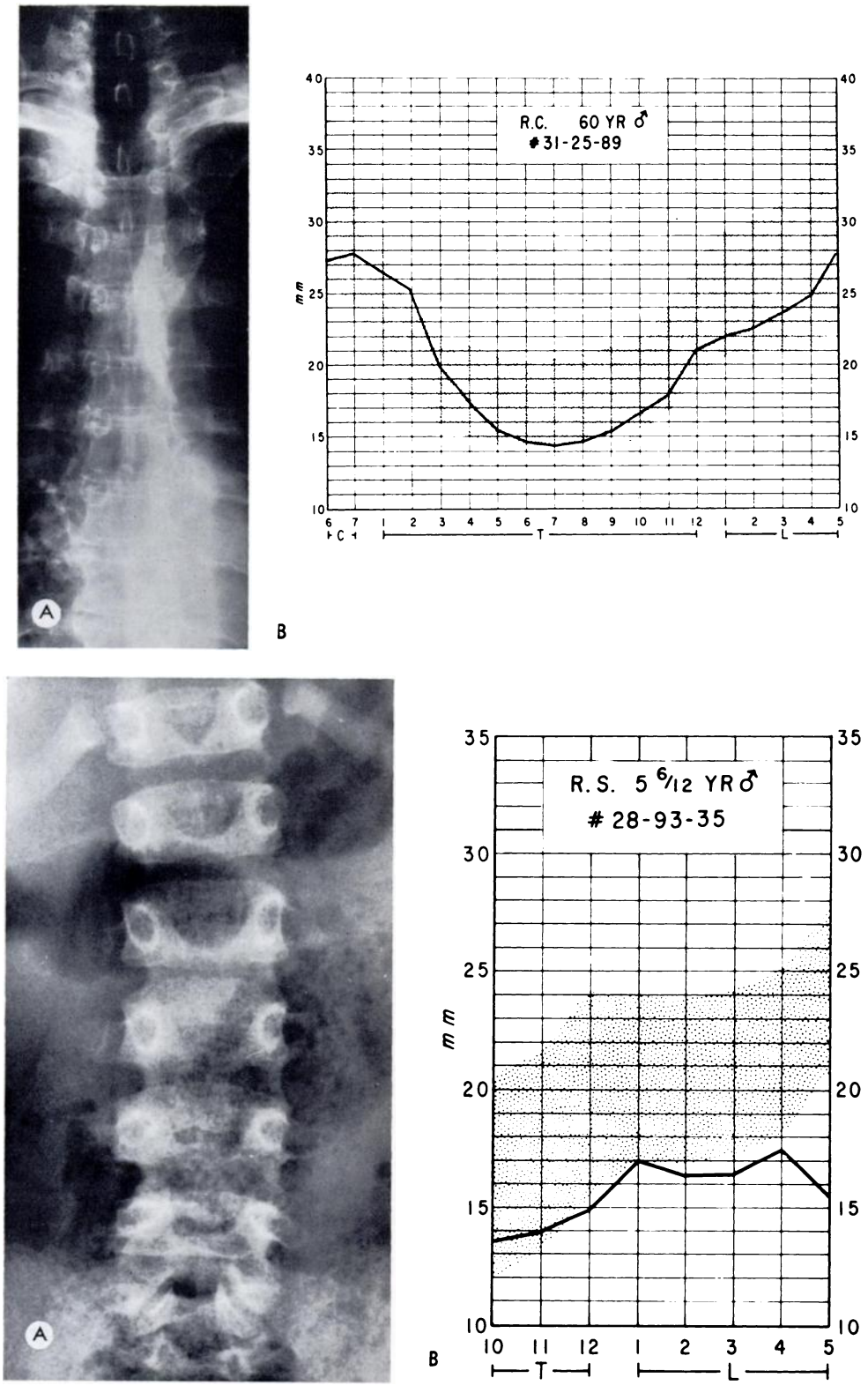


FIG. 6. Case II. (*A*) Diminished IPD's, more so in the lower lumbar and sacral spinal canal. (*B*) Graph depicts patient's measurements plotted against 90 per cent tolerance range for his age group.

FIG. 5. Case I. (*A*) Mild degenerative changes in middle and lower thoracic spine. Mid-thoracic interpediculate distances appear small. Vertebral morphology is otherwise normal. (*B*) Graph compares patient's IPD's with 90 per cent tolerance range of normal. There is stenosis of the spinal canal, especially between T5 and T10. T7 and T8 are at the lower limit of the tolerance range.

Downloaded from www.ajronline.org by 151.181.39.14 on 10/27/26 from IP address 151.181.39.14. Copyright ARRS. For personal use only; all rights reserved

along with the anteroposterior spine roentgenogram (Fig. 5*A*).

At myelography, the contrast column moved freely to the upper level of T12 which it passed only after an appreciable interval of arrest. Despite a steep head-down table tilt, the remainder of the transit through the thoracic spinal canal was very slow and hesitant. After the contrast medium had accumulated at the cervical level, the table was tilted to a near-erect position. The column travelled caudad with ease as far as T3, where it gradually narrowed and again began a slow movement through the rest of the thoracic canal. The spinal cord appeared to narrow excessively as it entered the region of stenosis. No defect was reported at T7–8 but a persistent extradural defect was noted at T12–L1 on the left side. A lateral spot roentgenogram showed a ventral impression of the column at the same level.

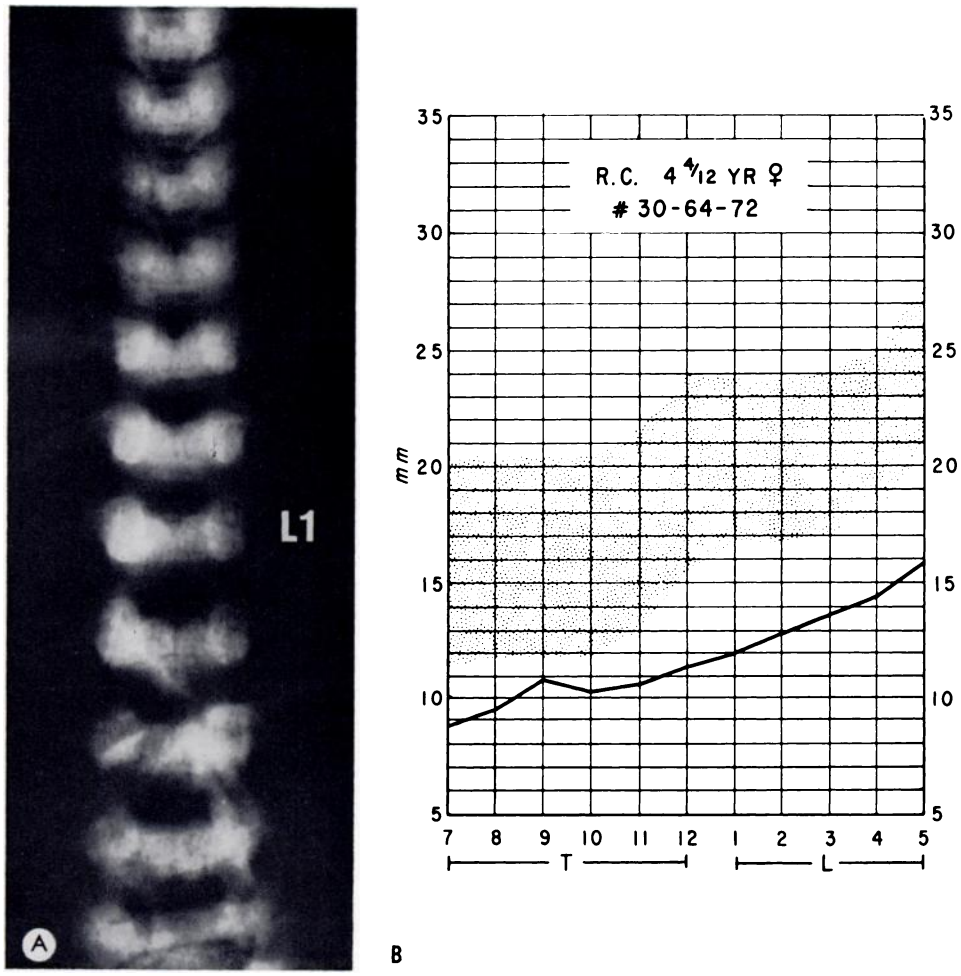


FIG. 7. Case III. (*A*) Massive vertebral bone structure due to osteopetrosis. Large pedicles have narrowed the IPD's. (*B*) Graph depicts patient's measurements and the 90 per cent tolerance range for her age group.

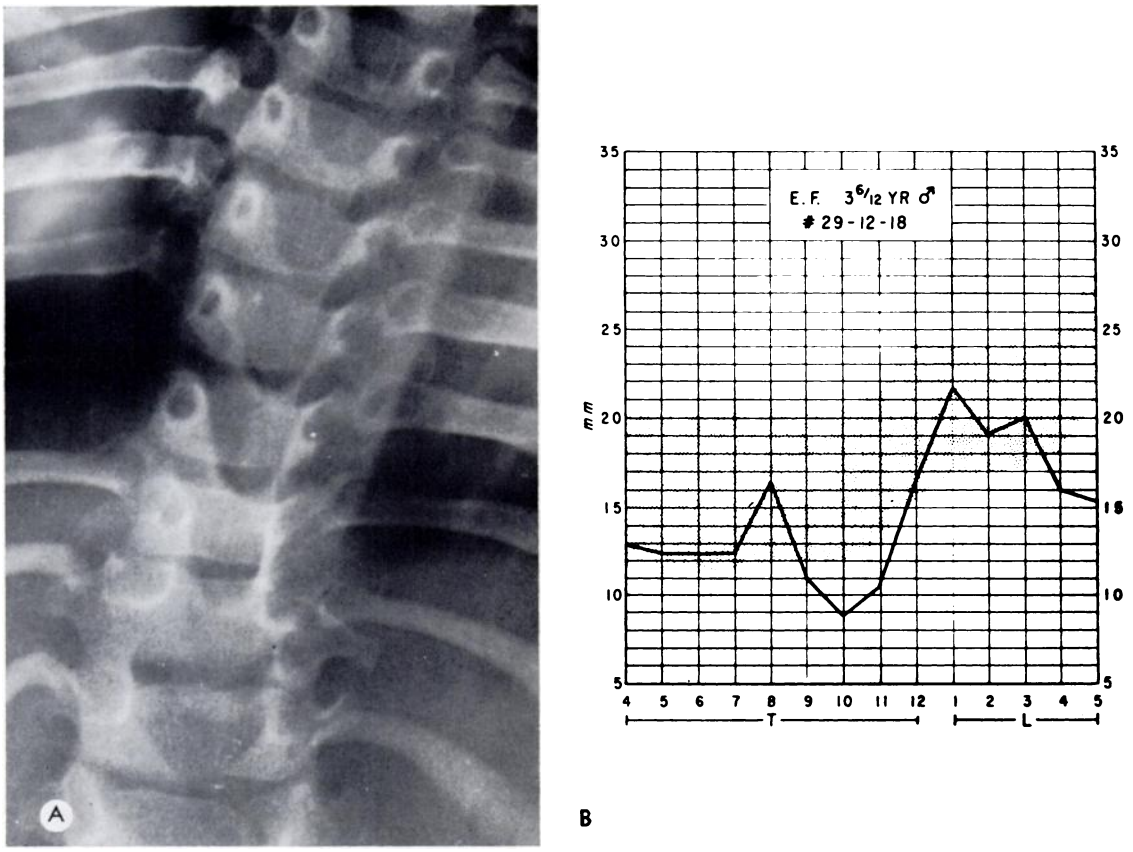


FIG. 8. Case IV. (*A*) Gross decrease of IPD measurements in lower thoracic spine. (*B*) Graph depicts patient's measurements and the 90 per cent tolerance range for his age group. Note variable size of vertebral bodies.

It was felt that the impression of thoracic spinal canal stenosis which had been gleaned from the conventional spine roentgenograms was amply supported both by the unusual dynamics seen at fluoroscopy and the unusual findings demonstrated on spot roentgenograms.

Comment. This patient demonstrated cord dysfunction roughly corresponding to the level of stenosis. It is suggested that he became symptomatic because degenerative changes developed in the thoracic spine leading to encroachment on the already narrow spinal canal and consequent spinal cord compression.

Despite lack of surgical confirmation, the presence of stenosis seems well established. The fact that the smallest IPD's are at, but not below, the tolerance limits does not militate against a diagnosis of stenosis. As stated previously in this paper, failure to exceed the limits cannot guarantee normalcy. The distribution curve of IPD measurements of patients with stenosis is not known, but evidently overlaps the distribution for normals. This patient's measurements fall within that area.

The following cases demonstrate more striking stenosis than Case I. Nonetheless, these children were asymptomatic. The lack of clinical manifestations probably depends on the greater ability of immature nervous tissue to compensate during the phase of active growth and maturation.

CASE II. R.S., a $5\frac{1}{2}$ year old male achondroplastic dwarf, was referred to this hospital for evaluation of mental retardation. He had no signs, symptoms or neurologic findings which might be ascribed to lumbar spinal canal stenosis. Anteroposterior lumbar spine roentgeno-

Downloaded from www.ajronline.org by 151.181.39.14 on 10/27/26 from IP address 151.181.39.14. Copyright ARRS. For personal use only; all rights reserved

Downloaded from www.ajronline.org by 151.181.39.14 on 10/27/26 from IP address 151.181.39.14. Copyright ARRS. For personal use only; all rights reserved

grams showed IPD's below T10 generally to be at or beyond the lower end of the 90 per cent tolerance range (Fig. 6, *A* and *B*). Stenosis was greatest at lower lumbar and sacral levels. Sagittal diameters of the spinal canal were also diminished.

CASE III. R.C., a 4 year old female, suffered from osteopetrosis. She was found to have stenosis of her spinal canal on roentgenologic examination, although there were no clinical findings related to it. Roentgenograms showed massive vertebral bone structure, especially in the pedicles. Enlargement of pedicles had caused decrease of IPD's (Fig. 7, *A* and *B*). Sagittal diameters of her lumbar spinal canal were also diminished.

CASE IV. E.F., a 3 year old male, suffered from osteochondrodystrophy, basilar invagination, partial atlanto-occipital assimilation, hemivertebra at T2 and ventral hemivertebra at T10. There were no clinical findings which could be related to stenosis of the spinal canal although roentgenograms showed gross decrease of IPD's in the lower thoracic spine. At the same level, there was marked increase of mid-sagittal diameter of the spinal canal. Roentgenographic abnormalities and a plot of the patient's IPD's are shown in Figure 8, *A* and *B*.

SUMMARY

In this article refined maximum and minimum interpediculate distances are published for use in diagnosis of intraspinal tumor and detection of spinal canal stenosis. The subject of spinal canal stenosis due to decrease of interpediculate distances is virtually unexplored. Availability of minimum norms should facilitate such studies.

Concerning spinal canal stenosis due to diminished interpediculate distances, it is apparent that (a) there is such an anomaly, (b) it may, but need not be, associated with diminished mid-sagittal diameter of the spinal canal, and (c) it may be symptomatic. Illustrative cases are presented.

Vincent C. Hinck, M.D.
Department of Radiology
University of Oregon Medical School
3181 S. W. Sam Jackson Park Road
Portland, Oregon 97201

Grateful acknowledgment is made for use of the Health Sciences Computing Facility, Center for the Health Sciences, University of California, Los Angeles.

REFERENCES

1. DIXON, W. J., and MASSEY, F. J., JR. Introduction to Statistical Analysis. McGraw-Hill Book Company, New York, 1957, p. 130.
2. ELSBERG, C. A., and DYKE, C. G. Diagnosis and localization of tumors of spinal cord by means of measurements made on x-ray films of vertebra, and correlation of clinical and x-ray findings. *Bull. Neurol. Inst. New York*, 1934, *3*, 359–394.
3. EPSTEIN, B. S., EPSTEIN, J. A., and LAVINE, L. Effect of anatomic variations in lumbar vertebrae and spinal canal on cauda equina and nerve root syndromes. AM. J. ROENTGENOL., RAD. THERAPY & NUCLEAR MED., 1964, *91*, 1055–1063.
4. HINCK, V. C., and HOPKINS, C. E. Sagittal diameter of lumbar spinal canal. In progress.
5. HINCK, V. C., and SACHDEV, N. S. Developmental stenosis of cervical spinal canal. To be published.
6. LANDMESSER, W. E., JR., and HEUBLEIN, G. W. Measurement of normal interpedicular space of child. *Connecticut M. J.*, 1953, *17*, 310–313.
7. SCHWARZ, G. S. Width of spinal canal in growing vertebra with special reference to sacrum: maximum interpediculate distances in adults and children. AM. J. ROENTGENOL., RAD. THERAPY & NUCLEAR MED., 1956, *76*, 476–481.
8. SILVERMAN, F. N. Personal communication.
9. SIMRIL, W. A., and THURSTON, D. Normal interpediculate space in spines of infants and children. *Radiology*, 1955, *54*, 340–347.
10. VERBIEST, H. Further experiences on pathological influence of developmental narrowness of bony lumbar vertebral canal. *J. Bone & Joint Surg.*, 1955, *37*-B, 576–583.